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Summer Research Project

Fineva: A Benchmark for Evaluating Finance-Focused LLM Agents

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Abstract

Large Language Models, including GPT-4, have shown considerable promise across diverse domains such as natural language processing, reasoning, medical applications, and financial analysis. Their ability to generate human-like text and process complex information makes them valuable tools in various applications, including text classification, sentiment analysis, and mathematical computations. Despite this, existing financial benchmarks like FinBen, PIXIU, and FinanceBench have limitations in evaluating the full spectrum of financial tasks, particularly those involving sophisticated decision-making and multi-faceted analyses. This research aims to address these gaps by developing a comprehensive evaluation framework for assessing LLM performance in real-world financial scenarios. Our framework incorporates diverse tasks and tools used by financial professionals, including real financial data from platforms like Bloomberg. The goal is to provide a thorough evaluation of LLM capabilities in handling core financial responsibilities and to enhance the understanding of their practical utility in the finance sector.

1 Introduction

Large Language Models (LLMs), such as GPT-4, have demonstrated significant potential in various fields, including general natural language processing tasks, reasoning, medical applications, ethics, education, natural and social sciences, agent applications, and other areas Chang et al. [2024]. Their capability to process and generate human-like text has rendered them suitable for tasks such as text classification Li et al. [2023], Rony et al. [2024], sentiment analysis Araci [2019], Zhang et al. [2023], and named entity recognition Wang et al. [2023]. Furthermore, LLMs have shown tremendous promise in completing code, synthesizing code from natural language descriptions Xu et al. [2022], and performing mathematical and statistical analysis Liu et al. [2023], making them invaluable assets in the finance industry. Their rapid response times and analytical capabilities are particularly advantageous in a field where timely and accurate information is crucial.

Existing financial domain evaluation benchmarks, such as FinBen [Xie et al., 2024], PIXIU [Xie et al., 2023], FinanceBench [Islam et al., 2023], and BizBench [Koncel-Kedziorski et al., 2023], have limitations in their evaluation tasks. These benchmarks primarily concentrate on a narrow range of Financial NLP tasks, such as information extraction and question answering, without adequately capturing the intricate and collaborative nature of real-world financial decision-making processes.

Among these benchmarks, FinBen offers the most detailed evaluation. However, it still lacks the ability to evaluate the diverse roles and fundamental skill sets required within the financial sector, as it does not consider the real tasks and corresponding skills needed. Making it is not possible to test the performance of LLMs in real financial world.

Benchmark	**Language**	**Task**	**RFD**	**RFS**	**TA**	**QA**	**AMC**	**RM**	**PP**
CFBenchmark	Chinese	7	x	x	3	x	x	x	x
Fin-Eva	Chinese	1	x	x	x	1	x	x	x
PIXIU	English	8	x	x	3	1	x	1	x
FinanceBench	English	1	x	x	x	1	x	x	x
BizBench	English	5	x	x	x	2	x	x	x
FinBen	English	24	x	x	8	3	x	4	x
Fineva	English	24	6	8	11	3	4	3	1

Table 1: Comparison of different financial benchmarks based on the number of tasks and datasets used, and the distribution across various aspects including Realtime Finance Data (RFD), Real Finance Scenario (RFS), Textual Analysis (TA), Question Answering (QA), Accounting Metric Calculation (AMC), Risk Management (RM), and Product Pricing (PP).

Currently, there is no standardized framework for evaluating the performance of LLMs (Large Language Models) in real-world financial practice. This research aims to bridge that gap by developing an evaluation framework that accurately assesses how well financial LLM agents perform core responsibilities within the industry. This issue is valuable because our research genuinely considers the tasks financial professionals encounter daily and the fundamental skills they require. By taking a broad perspective, we can analyze the effectiveness of LLMs in completing these real-world financial tasks. Additionally, we will incorporate real financial data, including tools like Bloomberg, which are commonly used by financial professionals, to better assess the performance of LLMs in actual environments and understand their impact in the meaningful field of finance.

In this project, we have made significant strides in developing a robust evaluation framework for Large Language Models (LLMs) within the financial sector. Our research has demonstrated that advanced models like GPT-4o and Claude excel in handling complex financial tasks, such as portfolio management and risk assessment, showing superior performance in these areas compared to earlier models like GPT-3.5. Our framework effectively bridges the gap left by existing benchmarks by incorporating a diverse range of financial scenarios and real-world data, such as that from Bloomberg. This comprehensive approach ensures a more accurate assessment of LLM capabilities, showcasing their potential to significantly enhance financial decision-making and analysis. Overall, our results reflect a high level of achievement and completeness in addressing the complexities of financial tasks and evaluating LLM performance.

2 Background

Our benchmark primarily evaluates the fundamental skills necessary for various financial positions, focusing on the core roles within investment banking. The job divisions are derived from the annual reports of Bulge Bracket banks[1], which are the most representative institutions in the industry. These banks have extensive business operations and set high industry standards. Their annual reports provide comprehensive data on revenue across different business divisions, detailed descriptions of departmental roles, and the latest organizational structures. The reports categorize these departments into front, middle, and back offices, adhering to financial industry terminology.

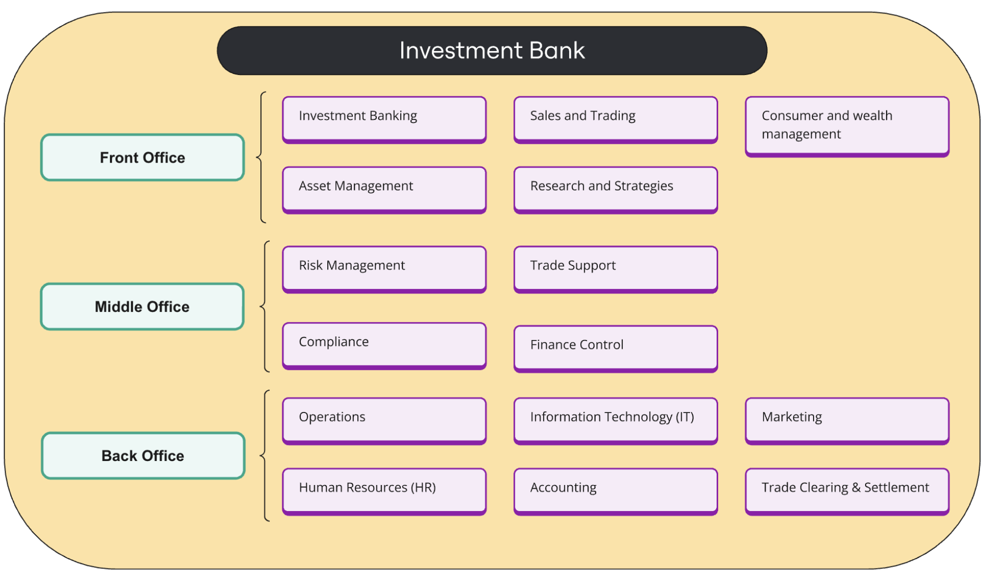

Figure 1: Structure of Investment Bank

[1]Bulge Bracket banks include: Goldman Sachs, JPMorgan Chase (J.P. Morgan), Morgan Stanley, Bank of America Merrill Lynch, Citigroup, Barclays, Deutsche Bank, and UBS.

Front Office: Typically client-facing and plays a key role as it has direct contact with customers, making it the primary revenue-generating part of the company. The front office includes divisions such as Investment Banking, Asset Management, Sales and Trading, Consumer and Wealth Management, and Research and Strategies.

Middle Office: Supports trading activities and includes units such as Risk Management, Compliance, Trade Support, and Finance Control.

Back Office: Handles administrative and operational tasks, including Operations, Human Resources (HR), Information Technology (IT), Accounting, Marketing, and Trade Clearing & Settlement.

2.1 Large Language Models in Finance

Large Language Models (LLMs) are increasingly employed in the finance sector for a range of applications. These models are particularly effective at tasks such as classifying news headlines, recognizing named entities, predicting stock movements, and providing basic financial analysis. LLMs excel in code generation, problem-solving, and statistical analysis, making them well-suited for the fast-paced and data-intensive nature of finance. Despite their advantages, there is currently no established standard for evaluating LLM performance in real-world financial scenarios.

2.2 Application in Non-Decision Making Scenarios

Large language models are transforming various aspects of financial operations. They are enhancing front-end marketing efforts, streamlining information collection and organization, and supporting back-end operational processes. Financial institutions are increasingly integrating these models into several domains, including payments, credit assessments, investment advice, research, and insurance.

Nvidia Survey

A recent survey by Nvidia, involving nearly 400 financial institutions, revealed that 43% are currently utilizing large models. The primary applications of these models include report generation (37%), optimizing customer experience and engagement (34%), generating synthetic data (33%), and marketing (32%).[2]

3 Fineva: Tasks, Skills and Data

We interviewed more than 20+ finance professionals working in investment banking, and reviewed job descriptions for both summer intern and full-time analyst positions in investment banking. From these, we summarized the scope of their daily work and the underlying skills required, which are presented in the table below.

3.1 Financial Modelling

Comparable Company Analysis

Real Financial Scenario: As an Investment Banking Analyst, it is essential to identify comparable companies by analyzing various financial indicators from different firms to evaluate a target company's position within the industry.

Data: The latest real financial data for 100 companies across the semiconductor industry's upstream, midstream, and downstream sectors was downloaded from Bloomberg. This data includes metrics such as EV/Sales, EV/EBIT, EV/EBITDA, P/E, PEG, P/B, and P/S ratios.

Test: Input the financial data of several companies within the same industry (with a specified target company) into an LLM to calculate these financial metrics. By analyzing the financial ratios of different companies within the same industry, the LLM can filter out the most suitable comparable companies.

[2]Source: Nvidia, "State of AI in Financial Services: 2024 Trends Survey Report", January 2024

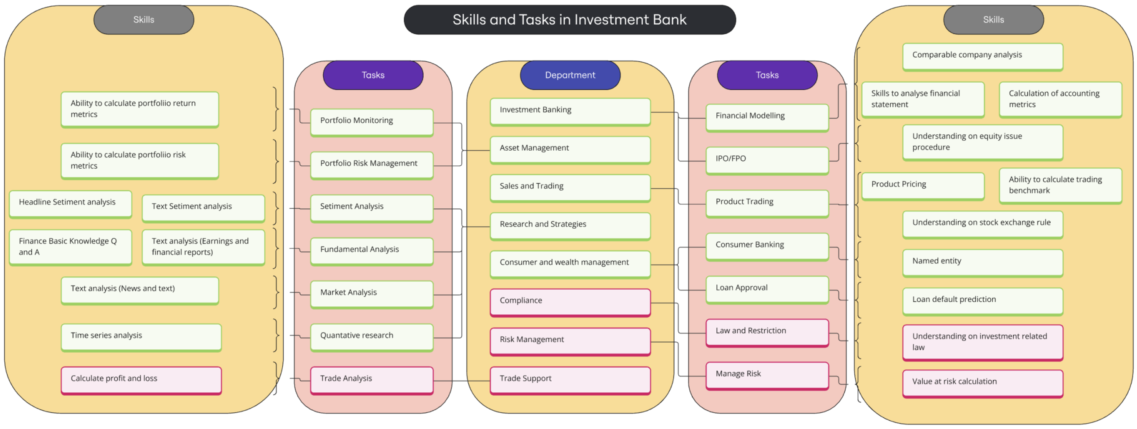

Figure 2: Skills and Tasks in Investment Banking

Skills to Analyze Financial Statements

Real Financial Scenario: As an Investment Banking Analyst, the ability to extract relevant tables and sections regarding a company's operating conditions from its annual report is crucial.

Data: The largest 150 publicly listed companies from the S&P 500 holdings were selected, and their annual reports were obtained through public channels. These reports include a basic company overview, a summary of accounting and business data, changes in equity and shareholder information, financial accounting reports, etc. Although the content of the reports is almost the same, the format is unstructured.

Test: Input the annual reports in PDF format into an LLM, instructing it to extract the most important financial tables, including the balance sheet, cash flow statement, and income statement.

Calculation of Accounting Metrics

Real Financial Scenario: As an Investment Banking Analyst, calculating a company's financial metrics from the balance sheet, cash flow statement, and income statement is essential for subsequent analysis.

Data: Fifteen years of real financial data for the US Magnificent 7 Stocks [Alphabet (GOOGL), Amazon (AMZN), Apple (AAPL), Meta Platforms (META), Microsoft (MSFT), NVIDIA (NVDA), and Tesla (TSLA)] were downloaded from Bloomberg. These data are the actual metrics that financial analysts typically review.

Test: Input the three financial statements into an LLM to calculate various financial metrics, including Current Ratio, Debt-to-Equity Ratio, Gross Margin, Net Profit Margin, Return on Assets (ROA), Return on Equity (ROE), EV/Sales, EV/EBIT, EV/EBITDA, P/E, PEG, P/B, and P/S ratios.

3.2 IPO/FPO

Understanding IPO Procedures

Real Financial Scenario: As an Investment Banking Analyst, it is essential to understand how to assist client companies in raising capital through a stock exchange.

Data: Procedures regarding IPOs were downloaded from the NYSE, HKEX, and Nasdaq, and 100 questions with standard answers were formulated based on this information. The questions include multiple-choice, short answer, and true/false formats.

Test: Input these questions into an LLM and have the LLM answer them.

3.3 Portfolio Monitoring

Ability to Calculate Portfolio Return Metrics

Real Financial Scenario: As a Portfolio Manager, it is necessary to understand the performance metrics of the constructed portfolio.

Data: Real stock data from Bloomberg was used, consisting of 15 stocks' prices over 30 days. Additionally, 100 different weighting combinations were randomly set as a testing set, with stocks labeled by industry and region.

Test: Provide the LLM with the 30-day prices, volatility data, regions, industries, and weightings for the 15 stocks, and instruct the LLM to calculate the portfolio's cumulative and daily returns, as well as the portfolio's weighting across different regions and industries.

3.4 Portfolio Risk Monitoring

Ability to Calculate Portfolio Risk Metrics

Real Financial Scenario: As a Portfolio Manager, it is crucial to understand the risk metrics of the constructed portfolio.

Data: Real stock data from Bloomberg was used, consisting of 15 stocks' prices and volatility over 30 days. Additionally, 100 different weighting combinations were randomly set as a testing set.

Test: Provide the LLM with the 30-day prices, volatility data, and weightings for the 15 stocks, and instruct the LLM to calculate the portfolio's Sharpe ratio, Sortino ratio, volatility, and Maximum Drawdown (MDD).

3.5 Product Trading

Product Pricing

Real Financial Scenario: As a Trader, it is important to know how to price different financial products and execute trading decisions.

Data: Real option data over 100 days was obtained, including Quote Date, Underlying Asset's Last Price, Expiration Date, Days to Expiration (DTE), Call Option Implied, Volatility (IV), Call Option Delta, Put Option Delta, Strike Price, etc.

Test: Provide the LLM with the daily option data and instruct it to calculate the bid-ask price of put and call options.

Ability to Calculate Trading Benchmarks

Real Financial Scenario: As a Trader, it is essential to know the various indicators calculated based on price and volatility in the market, which help traders identify the best trading prices and volumes.

Data: Real trading data from 150 S&P 500 companies over three months was obtained from Bloomberg, including price and trading volume.

Test: Provide the LLM with the price and volatility data and instruct it to calculate VWAP, standard deviation, moving average, KD, and ADX.

Understanding Stock Exchange Rules

Real Financial Scenario: As a Trader, it is crucial to know the trading rules of different exchanges.

Data: Rules regarding trading times and regulations were downloaded from the NYSE, HKEX, and Nasdaq, and questions in multiple-choice, short answer, and true/false formats were created.

Test: Input these questions into an LLM and have the LLM answer them based on the question type.

3.6 Sentiment Analysis

Text Sentiment Analysis

Real Financial Scenario: As a Research Analyst, it is crucial to analyze market sentiment regarding various assets and events, as investor behavior can often be inferred from their emotions.

Data1: The Financial News Statements and Headlines dataset contains 2,510 headlines from sources like Yahoo Finance, annotated by three financial experts. Each headline receives a sentiment score between -1 (very negative) and 1 (very positive), indicating its bearish or bullish nature.

Test: Provide the sentiment-laden text to an LLM and instruct it to determine whether the sentiment is positive or negative.

Data2: Microblog messages and News Statements and Headlines, annotated by experts, with sentiment scores ranging from -1 (very negative) to 1 (very positive), indicating bearish or bullish sentiment.

Test: Provide the sentiment-laden text to an LLM and instruct it to determine whether the sentiment is positive or negative.

Data3: News text with hawkish-dovish sentiment, annotated by experts with a sentiment score between -1 (very hawkish) and 1 (very dovish).

Test: Provide the sentiment-laden text to an LLM and instruct it to determine whether the sentiment is hawkish or dovish.

Headline Sentiment Analysis

Data: Political News Articles annotated by experts with sentiment scores ranging from -1 (very negative) to 1 (very positive), indicating bearish or bullish sentiment.

Test: Provide the sentiment-laden headlines to an LLM and instruct it to determine whether the sentiment is positive or negative.

3.7 Fundamental Analysis

Text Analysis (Earnings Call Transcripts)

Real Financial Scenario: As a Research Analyst, it is essential to quickly and accurately summarize long financial documents.

Data: ECTSum, a new dataset containing transcripts of earnings calls (ECTs) hosted by publicly traded companies, along with short, expert-written, telegram-style bullet point summaries derived from corresponding Reuters articles. ECTs are long, unstructured documents without any prescribed length or format.

Test: Provide the text to an LLM and, in the prompt, instruct the LLM to summarize the article in point form with no more than 10 bullet points, then evaluate the accuracy.

Text Analysis (Financial Reports)

Real Financial Scenario: As a Research Analyst, it's essential to quickly and accurately summarize various sections of long annual reports.

Data: FNS 2020 annual reports produced by UK firms listed on The London Stock Exchange (LSE). The task involves extracting summaries from different key sections typically referred to as "narrative sections" or "front-end" sections, containing textual information and reviews by the firm's management and board of directors, along with annotated summary points.

Test: Provide the text to an LLM and, in the prompt, instruct the LLM to summarize the article in point form with no more than 10 bullet points, then evaluate the accuracy.

Finance Basic Knowledge Q and A

Real Financial Scenario: As a Research Analyst, it's important to have a solid understanding of basic finance knowledge.

Data: CFA exam questions, covering topics such as professional ethics, financial statement analysis, quantitative methods, economics, fixed income securities analysis, equity securities analysis, portfolio management, and derivative financial instruments.

Test: Input the questions into an LLM and instruct the LLM to answer them based on the question type.

3.8 Market Analysis

Text Analysis (News and Extraction)

Real Financial Scenario: As a Research Analyst, it's essential to quickly and accurately summarize long news articles.

Data: The dataset includes 226,711 news articles, each paired with a brief one-sentence summary. These articles were sourced from BBC between 2010 and 2017, spanning various topics such as News, Politics, Sports, Weather, Business, Technology, Science, Health, Family, Education, Entertainment, and Arts. Annotated summary points are also provided.

Test: Submit the text to a language model, instructing it to generate a one-sentence summary of the article. Assess the model's accuracy in producing the summary.

3.9 Quantitative Research

Time Series Analysis

Real Financial Scenario: As a Quant Analyst, it's important to perform accurate machine learning model analysis on financial time series data.

Data: Data collected from CRSP, CompuStat, and WRDS Beta Suite, containing information such as:

- Ticker: the tickers for 30 stocks (current constituents of the DJIA index with DOW replaced by C). - Month: From December 2009 till November 2019, using covariates of the current month to predict the return of the next month. - Covariates: A total of 49 covariates, including current-month return, measures constructed using price data, beta, alpha, idiosyncratic, and total volatility of a stock, as well as financial ratios. Further explanations are provided in "explanation of some variables.xlsx" and "WRDS Industry Financial Ratio Manual.pdf."

Test: Provide the data to an LLM and, in the prompt, instruct the LLM to perform [decision tree, clustering, regression] analysis, then evaluate the accuracy.

3.10 Risk Management

Value at Risk Calculation

Real Financial Scenario: As a risk analyst, it's crucial to understand the risk data of your stock.

Data: Price data for the largest 150 publicly traded companies in the S&P 500 over 30 days, sourced from Bloomberg.

Test: Provide the 30-day stock price data to an LLM and instruct it to calculate the stock's value at risk.

3.11 Consumer Banking

Named Entity Recognition

Real Financial Scenario: As a consumer banke, it's important to identify Named Entity.

Data: Extract entities like LOCATION, ORGANIZATION, and PERSON from financial agreements and SEC filings, using the NER (Alvarado et al., 2015) and FINER-ORD (Shah et al., 2023b) datasets.

Test: Input the text into an LLM to see if it can successfully identify named entities within the content.

3.12 Loan Approval

Loan Default Prediction

Real Financial Scenario: As a consumer banker, it's necessary to analyze each customer's likelihood of default to approve loans.

Data: The dataset contains information on default payments, demographic factors, credit data, payment history, and bill statements of credit card clients in Taiwan from April 2005 to September 2005.

Test: Provide the dataset to an LLM and instruct it to predict the likelihood of a client defaulting.

3.13 Law and Restriction

Understanding Investment-Related Law

Real Financial Scenario: As a member of the finance compliance department, it's essential to interpret and apply complex regulatory frameworks in real-world scenarios.

Data: Questions related to the Code of Conduct for Persons Licensed by or Registered with the Securities and Futures Commission; Basel III, US, and UK government regulations.

Test: Input the questions into an LLM and instruct it to answer them based on the question type.

3.14 Risk Management

Value at Risk Calculation

Real Financial Scenario: As a trade support specialist, it's important to understand the performance data of your product after a day of trading.

Data: Real stock data from Bloomberg, consisting of the opening and closing prices of 50 stocks over one day, with 100 different randomly assigned opening and closing holdings as the testing set.

Test: Provide the 30-day price data, volatility data, region, and industry weighting of 15 stocks to an LLM and instruct it to calculate the portfolio's cumulative and daily returns, as well as the portfolio's weighting across different regions and industries.

4 Evaluation

This project selected three of the most common and accessible large language models (LLMs) available today to evaluate their performance across various financial tasks using our custom benchmark. These models are widely recognized for their importance in the field of artificial intelligence and natural language processing:

1. **GPT-3.5**: Developed by OpenAI, GPT-3.5 is a prominent language model known for its strong performance in a wide range of natural language processing tasks. Its versatility and broad application make it a critical tool for exploring the capabilities of LLMs in different domains, including finance.
2. **GPT-4o**: As an advanced iteration of OpenAI's GPT-4 model, GPT-4o offers enhanced accuracy and deeper understanding of complex tasks. This model is significant for pushing the boundaries of what LLMs can achieve, especially in specialized areas such as financial analysis and decision-making.
3. **Claude**: Developed by Anthropic, Claude is designed with a focus on safety and interpretability. Its optimization for ethical considerations and responsible AI usage makes it particularly relevant in fields like finance, where the implications of AI-driven decisions are profound.

These models were chosen not only because of their widespread availability but also due to their impact on the AI landscape. Their integration into the financial sector has been the subject of increasing interest. Researchers have explored their potential applications, opportunities, and challenges in finance research and decision-making processes.

4.1 Experimental Settings

Each model was evaluated on 24 distinct financial tasks, ranging from fundamental analysis to complex portfolio metrics calculations. The tasks include a variety of real-world financial scenarios such as comparable company analysis, portfolio return, and risk metrics calculation, as well as sentiment analysis of financial texts. Datasets marked with (*) are constructed by us.

Data	Task	Test	Evaluation	License
CompCorp(*)	Comparable Company Analysis	150	F1, Accuracy	Bloomberg
FinStatm(*)	Skills to Analyze Financial Statements	150	F1, Accuracy	Public
AcctCal(*)	Calculation of Accounting Metrics	147	F1, Accuracy	Public
IPO(*)	Understanding IPO Procedures	300	F1, Accuracy	Public
Prtn(*)	Ability to Calculate Portfolio Return Metrics	150	F1, Accuracy	Bloomberg
Prisk(*)	Ability to Calculate Portfolio Risk Metrics	150	F1, Accuracy	Bloomberg
AAPL Option Chains	Product Pricing	500	F1, Accuracy	CC0: Public Domain
TradBen(*)	Ability to calculate trading benchmark	150	F1, Accuracy	Bloomberg
ExcRule(*)	Understanding on stock exchange rule	500	F1, Accuracy	Public
FiGSA Ruan et al. [2022]	Text Sentiment analysis	2,510	F1, Accuracy	CC BY-NC 4.0
FOMC Shah et al. [2023a]	Text Sentiment analysis	496	F1, Accuracy	CC BY-NC 4.0
TSA Cortis et al. [2017]	Text Sentiment analysis	561	F1, Accuracy	CC BY-NC-SA 4.0
NewsMTSC Hamborg and Donnay [2021]	Headline Sentiment analysis	1130	F1, Accuracy	Public
ECTSum Mukherjee et al. [2022]	Text analysis (Earnings Call Transcripts, financial reports)	1,684	F1, Accuracy	GPL-3.0
FNS 2020 Mukwazvure and Supreethi [2015]	Text analysis (Earnings Call Transcripts, financial reports)	50,000	F1, Accuracy	CC BY 4.0
Flare-cfa	Finance Basic Knowledge QnA	1032	F1, Accuracy	Public
XSum Narayan et al. [2018]	Text analysis (News and extraction)	226,711	F1, Accuracy	MIT License
WRDS Beta Suite	Time series analysis	3600	F1, Accuracy, AUC	WRDS
Parcel(*)	Value at risk calculation	150	F1, Accuracy	Bloomberg
NER Alvarado et al. [2015]	Named entity	980	Entity F1	CC BY-SA 3.0
FiNER-ORD Shah et al. [2023b]	Named entity	1080	Entity F1	CC BY-NC 4.0
Default of Credit Card Clients Dataset	Loan Default prediction	30,000	F1, Accuracy, AUC	CC0: Public Domain
Compliance(*)	Understanding on investment related law	500	F1, Accuracy	Public
PnLcal(*)	Calculate profit and loss	150	F1, Accuracy	Bloomberg

Table 2: Data, Tasks, Tests, Evaluations, and Licenses

The evaluation was conducted using a standardized environment with consistent computational resources to ensure fairness in comparison. All tasks were performed in a zero-shot manner, with no task-specific fine-tuning applied to any model. The results highlight the capabilities and limitations of each LLM in handling various financial tasks, providing insights into their practical utility in the financial industry.

5 Result Evaluation

Data	Task	GPT-3.5	GPT-4o	Claude
CompCorp(*)	Comparable Company Analysis	1	1	1
AcctCal(*)	Calculation of Accounting Metrics	1	1	1
IPO(*)	Understanding IPO Procedures	0.717	0.5	0.761
Prtn(*)	Ability to Calculate Portfolio Return Metrics	0.102	0.82	0.76
Prisk(*)	Ability to Calculate Portfolio Risk Metrics	0.125	0.92	0.83
AAPL Option Chains	Product Pricing	0.121	1	1
TradBen(*)	Ability to calculate trading benchmark	0.343	0.862	0.853
ExcRule(*)	Understanding on stock exchange rule	0.685	0.29	0.692
FiGSA (Ruan et al., 2022)	Text Sentiment analysis	0.685	0.635	0.685
FOMC (Shah et al., 2023)	Text Sentiment analysis	0.661	0.708	0.72
NewsMTSC (Hamborg & Donnay, EACL 2021)	Headline Sentiment analysis	0.692	0.788	0.769
ECTSum (Mukherjee et al., 2022)	Text analysis (Earnings Call Transcripts, financial reports)	0.542	0.896	0.77
FNS 2020 (Mukwazvure & Supreethi, 2015)	Text analysis (Earnings Call Transcripts, financial reports)	0.694	0.823	0.858
Flare-cfa	Finance Basic Knowledge QnA	0.62	0.86	0.90
XSum: (Narayan et al., EMNLP 2018)	Text analysis (News and extraction)	0.153	0.134	0.156
Parcel(*)	Value at risk calculation	0.89	1	1
NER (Alvarado et al., 2015)	Named entity	0.208	0.272	0.213
Compliance(*)	Understanding on investment related law	0.492	0.57	0.62
PnLcal(*)	Calculate profit and loss	1	1	1

Table 3: Comparison of GPT-3.5, GPT-4o, and Claude across various financial tasks.

5.1 Comparable Company Analysis & Accounting Metrics Calculation

Across tasks related to **Comparable Company Analysis** and **Accounting Metrics Calculation**, all models (GPT-3.5, GPT-4o, and Claude) demonstrate consistent performance, achieving the highest possible scores. This indicates a strong capability in handling fundamental financial analysis tasks, particularly those requiring structured numerical computations and financial statement analysis. The consistency across models in these areas highlights their reliability in supporting standard financial processes and their ability to interpret and apply key financial metrics accurately.

5.2 Portfolio Management & Risk Assessment

In the domain of **Portfolio Management** and **Risk Assessment**, including tasks such as **Portfolio Return and Risk Metrics Calculation**, the performance of the models diverges significantly. GPT-4o and Claude excel in these tasks, indicating their advanced abilities in processing and analyzing complex financial data. In contrast, GPT-3.5 struggles with these tasks, especially in portfolio return metrics, where its performance is notably lower. This suggests that while newer models like GPT-4o and Claude can handle the intricacies of portfolio analysis, earlier models may still face challenges, particularly in scenarios requiring nuanced understanding and processing of financial risk and return metrics.

5.3 Product Pricing & Benchmark Calculation

For **Product Pricing** and **Benchmark Calculation** tasks, such as those involving **AAPL Option Chains** and **Trading Benchmark Calculations**, GPT-4o and Claude again demonstrate superior performance, achieving perfect scores. GPT-3.5, while generally competent, shows some limitations in benchmark calculation tasks. This pattern underscores the advanced capabilities of the latest models in accurately pricing complex financial products and calculating trading benchmarks, which are essential for effective financial decision-making and strategy development.

5.4 Text and Sentiment Analysis

In **Text and Sentiment Analysis** tasks, models exhibit varying degrees of proficiency. Claude shows strong performance across most datasets, particularly in handling tasks like **Headline Sentiment Analysis** and **Text Analysis of Earnings Call Transcripts**. GPT-4o also performs well, although its scores are slightly lower than Claude's in some instances. GPT-3.5, while competent, trails behind in this category, particularly in nuanced text analysis tasks. This suggests that while GPT-4o and Claude are more adept at processing and interpreting financial texts, especially in sentiment-heavy contexts, GPT-3.5 may struggle with the complexity and subtleties required for high-accuracy sentiment analysis.

5.5 Compliance & Law Understanding

For tasks related to **Compliance** and **Investment Law Understanding**, the models display varied performance. Claude leads slightly, reflecting its broader knowledge base and advanced reasoning capabilities, particularly in interpreting investment-related laws. This differentiation highlights the varying strengths of each model, with newer models like GPT-4o and Claude being better equipped for tasks requiring comprehensive understanding and application of financial regulations.

5.6 Profit/Loss Calculations

For **Profit and Loss Calculations**, all models, including GPT-3.5, achieve high scores, indicating their strong numerical and computational abilities in this area. This reflects the models' reliability in performing essential financial calculations and their competence in supporting financial decision-making processes.

5.7 Overall Performance

Overall, the evaluation reveals a clear distinction between earlier models like GPT-3.5 and more advanced ones like GPT-4o and Claude. While GPT-3.5 is reliable for basic financial calculations and structured tasks, it faces challenges in more complex scenarios involving nuanced analysis and multi-layered reasoning, particularly in portfolio management, risk assessment, and text analysis. On the other hand, GPT-4o and Claude consistently excel across a wide range of tasks, showcasing their enhanced capabilities in handling both quantitative and qualitative financial tasks. This highlights the importance of model choice depending on the complexity of the financial task at hand, with newer models offering substantial improvements in accuracy and reasoning abilities.

6 Future Works

The integration of Auto-GPT and prompt engineering into the financial sector presents significant opportunities for advancing the complexity and accuracy of tasks such as stock price prediction, complex pricing models, and financial analysis report generation. Auto-GPT's capabilities in processing extensive datasets and generating contextually appropriate outputs position it as a transformative tool for financial analysis and decision-making. By employing prompt engineering, financial professionals can calibrate the AI to meet specific analytical requirements, thus enhancing the precision and relevance of the results. The synergy between Auto-GPT and prompt engineering has the potential to revolutionize the financial industry, offering more sophisticated predictive models, comprehensive analytical frameworks, and efficient report generation processes. This advancement could lead to more informed and strategic investment decisions, underscoring the importance of these technologies in the evolving landscape of finance.

7 Conclusion

The development of a comprehensive evaluation framework for Large Language Models (LLMs) in the financial domain addresses a significant gap in current benchmarks, which often fail to capture the complexity of real-world financial tasks. By focusing on a broad range of financial activities, including portfolio management, risk assessment, and fundamental analysis, our framework provides a more accurate measure of LLM performance in the finance industry. The results demonstrate that advanced models like GPT-4o and Claude exhibit strong capabilities in handling intricate financial analyses, while earlier models like GPT-3.5 show limitations in more complex tasks. This research highlights the potential of LLMs to support and enhance financial decision-making processes but also underscores the need for continued development and refinement of evaluation methodologies to fully leverage their capabilities in practical settings.

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Appendix - Sample Prompt

Sample Prompt - Accounting Metric Calculation

Using the provided Balance Sheet, Income Statement, and Cash Flow Statement, calculate the following ratios:

- **Current Ratio**: [ans]
- **Debt-to-Equity Ratio**: [ans]
- **Gross Margin**: [ans]
- **Net Profit Margin**: [ans]
- **Return on Assets (ROA)**: [ans]
- **Return on Equity (ROE)**: [ans]

Please provide the answers in the format specified above at the end.

Sample Prompt - Portfolio Metrics

Using the provided portfolio and their respective weightings on the last day, calculate the following:

- **Return**: [ans]
- **Sector Breakdown**: [ans]

Please provide the answers in the format specified above at the end.

Sample Prompt - Portfolio Risk Metrics

Summarize the provided text in point form with less than 10 points.

Using the provided portfolio and their respective weightings on the last day, calculate the following:

- **Value at Risk (VaR) 99%**: [ans]
- **Value at Risk (VaR) 95%**: [ans]
- **Volatility**: [ans]
- **Sharpe Ratio**: [ans]
- **Sortino Ratio**: [ans]

Please provide the answers in the format specified above at the end.

Sample Prompt - Named Entity

In the sentences extracted from financial agreements in U.S. SEC filings, identify the named entities that represent a person ('PER'), an organization ('ORG'), or a location ('LOC'). Please identify and output all the entity in format (entity name, entity type).

ans

Please provide the answers in the following format at the end.

Sample Prompt - Option Pricing

Based on the provided data, calculate the option price using both the Black-Scholes (BS) model and the binomial tree method:

- **BS Model**: [ans]
- **Binomial Tree**: [ans]

Please provide the answers in the format specified above at the end.

Sample Prompt - Sentiment Analysis

Based on the provided text, give a sentiment score for the mentioned target. (ans should be 'HAWKISH', 'DOVISH', or 'NEUTRAL')

- **Sentiment Score**: [ans]

Please provide the answers in the format specified above at the end.

Sample Prompt - Sentiment Analysis_2

Based on the provided text, give a sentiment score for the mentioned target. (ans should be "NEGATIVE", "NEUTRAL" or "POSITIVE")

- **Sentiment Score**: [ans]

Please provide the answers in the following format at the end.

Sample Prompt - Q and A (MC)

Answer the provided multiple choice question, and output your choice.

ans

Please provide the answers in the following format at the end.

Sample Prompt - Q and A (Short Question)

Answer the provided Short question, and output your answer.

ans

Please provide the answers in the following format at the end.

Sample Prompt - Calculate Trading Benchmarks

Using the provided stock's daily price and trading volatility, calculate the VWAP, standard deviation, moving average, KD, and ADX.

- **VWAP**: [ans]
- **Standard deviation** : [ans]
- **Moving average**: [ans]
- **KD**: [ans]
- **ADX**: [ans]

Please provide the answers in the following format at the end.

VeriGuide - Originality Report
Individual Report

Background Information

Submission Reference ID:	XXXXXX
School / Institution:	Faculty of Engineering, CUHK
Year:	2023
Term:	2
Course:	ENGG-0001-A
Title:	Summer Research Internship 2024
Assignment Number:	2
Assignment Marker:	CHUNG, Katie
Student:	XXXXXX
Student's School ID:	XXXXXX@link.cuhk.edu.hk
File Name:	XXXXXX_Final Report.pdf
Submitted on:	2024-08-13 17:50:13+0800

Similarity Statistics Overview

Similarity (at default settings):	7.91%
Similarity (at current filter settings):	7.91%
Current filter settings:	Leniency: Default Minimum number of meaningful words in a sentence: 5 (default) Remove Reference Entries (Beta Testing, support English only): Off (default) Some sources are excluded by user: No
Sentence(s) Selected By User To Export:	17

Similarity Details

Index:	**1**
Suspected Sentence:	Furthermore, LLMs have shown tremendous promise in completing code, synthesizing code from natural language descriptions Xu et al.
Source Content:	8 * 130 TFLOP/s * 6 * 7 * 24 * 3600 * 0.3 (utilization) ~= 1.1e21",6 weeks,"Large language models (LMs) of code have recently shown tremendous promise in completing code and synthesizing code from natural language descriptions.
Source:	https://epochai.org/data/epochdb/notable_ai_models.csv
From:	Internet

Index:	**2**

Suspected Sentence:	Data: ECTSum, a new dataset containing transcripts of earnings calls (ECTs) hosted by publicly traded companies, along with short, expert-written, telegram-style bullet point summaries derived from corresponding Reuters articles.
Source Content:	In this work, we present ECTSum, a new dataset with transcripts of earnings calls (ECTs), hosted by publicly traded companies, as documents, and experts-written short telegram-style bullet point summaries derived from corresponding Reuters articles.
Source:	https://rajdeep345.github.io/files/pdf/research/c09.pdf
From:	Internet

Index:	**3**
Suspected Sentence:	ECTs are long, unstructured documents without any prescribed length or format.
Source Content:	ECTs are long unstructured documents without any prescribed length limit or format.
Source:	https://rajdeep345.github.io/files/pdf/research/c09.pdf
From:	Internet

Index:	**4**
Suspected Sentence:	Existing financial domain evaluation benchmarks, such as FinBen [Xie et al.
Source Content:	Existing financial domain evaluation benchmarks including FLUE Shah et al.
Source:	https://arxiv.org/html/2402.12659v1
From:	Internet

Index:	**5**
Suspected Sentence:	Existing financial domain evaluation benchmarks, such as FinBen [Xie et al.
Source Content:	In this work, we introduce a comprehensive financial benchmark the FinBen specifically designed for evaluating LLMs in the financial domain.
Source:	https://arxiv.org/html/2402.12659v1
From:	Internet

Index:	**6**
Suspected Sentence:	Each headline receives a sentiment score between -1 (very negative) and 1 (very positive), indicating its bearish or bullish nature.
Source Content:	“Given the following financial text, return a sentiment score for Ashtead as a floating-point number ranging from -1 (indicating a very negative or bearish sentiment) to 1 (indicating a very positive or bullish sentiment), with 0 designating neutral sentiment.
Source:	https://arxiv.org/html/2402.12659v1
From:	Internet

Index:	**7**
Suspected Sentence:	Data: Extract entities like LOCATION, ORGANIZATION, and PERSON from financial agreements and SEC filings, using the NER (Alvarado et al.
Source Content:	1) Named entity recognition extracts entities like LOCATION, ORGANIZATION, and PERSON from financial agreements and SEC filings, using the NER Alvarado et al.
Source:	https://arxiv.org/html/2402.12659v1
From:	Internet

Index:	**8**
Suspected Sentence:	SEC filings, identify the named entities that represent a person ('PER'), an organization ('ORG'), or a location ('LOC').
Source Content:	1) Named entity recognition extracts entities like LOCATION, ORGANIZATION, and PERSON from financial agreements and SEC filings, using the NER Alvarado et al.
Source:	https://arxiv.org/html/2402.12659v1
From:	Internet

Index:	**9**
Suspected Sentence:	SEC filings, identify the named entities that represent a person ('PER'), an organization ('ORG'), or a location ('LOC').
Source Content:	SEC filings, identify the named entities that represent a person ('PER'), an organization ('ORG'), or a location ('LOC').
Source:	https://arxiv.org/html/2402.12659v1
From:	Internet

Index:	**10**
Suspected Sentence:	Data: CFA exam questions, covering topics such as professional ethics, financial statement analysis, quantitative methods, economics, fixed income securities analysis, equity securities analysis, portfolio management, and derivative financial instruments.
Source Content:	The CFA dataset comprises Level 1 and Level 2 examination data, covering a wide range of topics such as ethics and professional standards, quantitative methods, economics, financial reporting and analysis, corporate finance, equity investments, fixed income, derivatives, alternative investments, and portfolio management.
Source:	https://arxiv.org/html/2407.00365v1
From:	Internet

Index:	**11**
Suspected Sentence:	Morgan), Morgan Stanley, Bank of America Merrill Lynch, Citigroup, Barclays, Deutsche Bank, and UBS.
Source Content:	• Bank of America Merrill Lynch • Barclays • Blackstone • Citi • Credit Suisse • Deutsche Bank • Goldman Sachs • HSBC • JPMorgan Chase • Morgan Stanley • UBS
Source:	https://www.investopedia.com/terms/b/bulgebracket.asp

From:	Internet

Index:	**12**
Suspected Sentence:	Data: FNS 2020 annual reports produced by UK firms listed on The London Stock Exchange (LSE).
Source Content:	For FNS 2020 task we focus on annual reports produced by UK firms listed on The London Stock Exchange (LSE).
Source:	http://wp.lancs.ac.uk/cfie/fns2020/
From:	Internet

Index:	**13**
Suspected Sentence:	The questions include multiple-choice, short answer, and true/false formats.
Source Content:	There are 5 types of questions that can be found in regular OWL assignments: true/false; multiple choice; multiple, multiple choice; matching; and short answer.
Source:	https://owl.oit.umass.edu/owlhelp/website/studentmanual/types_of_questions.htm
From:	Internet

Index:	**14**
Suspected Sentence:	- Covariates: A total of 49 covariates, including current-month return, measures constructed using price data, beta, alpha, idiosyncratic, and total volatility of a stock, as well as financial ratios.
Source Content:	Covariates: There are 49 covariates in total, including the current-month return and mea- sures constructed using the price data, and beta, alpha, idiosyncratic, and total volatility of a stock as well as financial ratios.
Source:	http://ass.3daixie.com/202404265864708281.html
From:	Internet

Index:	**15**
Suspected Sentence:	Further explanations are provided in “explanation of some variables.xlsx” and “WRDS Industry Financial Ratio Manual.pdf.
Source Content:	and “WRDS Industry Financial Ratio Manual.pdf” for explanations .
Source:	http://ass.3daixie.com/202404265864708281.html
From:	Internet

Index:	**16**
Suspected Sentence:	Making it is not possible to test the performance of LLMs in real financial world.
Source Content:	The effectiveness of LLMs in financial tasks is quantitatively assessed by comparing their recommendations against real-world financial data and historical market performance.
Source:	https://arxiv.org/html/2401.11641v1
From:	Internet

Index:	**17**
Suspected Sentence:	All tasks were performed in a zero-shot manner, with no task-specific fine10 tuning applied to any model.
Source Content:	Moreover, the zero-shot and few-shot learning of GPT models proved insufficient for accurate predictions in many cases, emphasizing the need for further fine-tuning on task-specific data.
Source:	https://aclanthology.org/2023.paclic-1.89.pdf
From:	Internet

Index:	**18**
Suspected Sentence:	Data: ECTSum, a new dataset containing transcripts of earnings calls (ECTs) hosted by publicly traded companies, along with short, expert-written, telegram-style bullet point summaries derived from corresponding Reuters articles.
Source Content:	In this work, we present ECTSum, a new dataset with transcripts of earnings calls (ECTs), hosted by publicly traded companies, as documents, and short experts-written telegram-style bullet point summaries derived from corresponding Reuters articles.
Source:	https://arxiv.org/abs/2210.12467
From:	Internet

Index:	**19**
Suspected Sentence:	ECTs are long, unstructured documents without any prescribed length or format.
Source Content:	ECTs are long unstructured documents without any prescribed length limit or format.
Source:	https://arxiv.org/abs/2210.12467
From:	Internet

Index:	**20**
Suspected Sentence:	Existing financial domain evaluation benchmarks, such as FinBen [Xie et al.
Source Content:	Existing financial domain evaluation benchmarks including PIXIU (Xie et al.
Source:	https://arxiv.org/pdf/2402.12659
From:	Internet

Index:	**21**
Suspected Sentence:	Existing financial domain evaluation benchmarks, such as FinBen [Xie et al.
Source Content:	5 Conclusion In this work, we present FinBen, a comprehensive benchmark specifically designed to evaluate LLMs in the financial domain.
Source:	https://arxiv.org/pdf/2402.12659
From:	Internet

Index:	**22**
Suspected Sentence:	Existing financial domain evaluation benchmarks, such as FinBen [Xie et al.
Source Content:	This breadth of coverage sets FinBen apart from existing financial benchmarks, enabling a more robust and nuanced evaluation of LLM capabilities.
Source:	https://arxiv.org/pdf/2402.12659
From:	Internet

Index:	**23**
Suspected Sentence:	Each headline receives a sentiment score between -1 (very negative) and 1 (very positive), indicating its bearish or bullish nature.
Source Content:	TSA “Given the following financial text, return a sentiment score for Ashtead as a floating-point number ranging from -1 (indicating a very negative or bearish sentiment) to 1 (indicating a very positive or bullish sentiment), with 0 designating neutral sentiment.
Source:	https://arxiv.org/pdf/2402.12659
From:	Internet

Index:	**24**
Suspected Sentence:	Data: Extract entities like LOCATION, ORGANIZATION, and PERSON from financial agreements and SEC filings, using the NER (Alvarado et al.
Source Content:	1) Named entity recognition extracts entities like LOCATION, ORGANIZATION, and PERSON from financial agreements and SEC filings, using the NER (Alvarado et al.
Source:	https://arxiv.org/pdf/2402.12659
From:	Internet

Index:	**25**
Suspected Sentence:	[2023a] Text Sentiment analysis 496 F1, Accuracy CC BY-NC 4.0 TSA Cortis et al.
Source Content:	, 2018) sentiment analysis 235 F1 Public TSA (Cortis et al.
Source:	https://arxiv.org/pdf/2402.12659
From:	Internet

Index:	**26**
Suspected Sentence:	[2015] Named entity 980 Entity F1 CC BY-SA 3.0 FiNER-ORD Shah et al.
Source Content:	, 2015) named entity recognition 980 Entity F1 CC BY-SA 3.0 FiNER-ORD (Shah et al.
Source:	https://arxiv.org/pdf/2402.12659
From:	Internet

Index:	**27**
Suspected Sentence:	SEC filings, identify the named entities that represent a person ('PER'), an organization ('ORG'), or a location ('LOC').
Source Content:	1) Named entity recognition extracts entities like LOCATION, ORGANIZATION, and PERSON from financial agreements and SEC filings, using the NER (Alvarado et al.
Source:	https://arxiv.org/pdf/2402.12659
From:	Internet

Index:	**28**
Suspected Sentence:	SEC filings, identify the named entities that represent a person ('PER'), an organization ('ORG'), or a location ('LOC').
Source Content:	SEC filings, identify the named entities that represent a person ('PER'), an organization ('ORG'), or a location ('LOC').
Source:	https://arxiv.org/pdf/2402.12659
From:	Internet